

ASHIRWAD KUMAR

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Graphic Era Deemed to be University, Dehradun

2023 – 2027

B.Tech in Computer Science & Engineering CGPA: 8.44/10

Class 12th **86.4%**

2021 – 2022

Class 10th **90.2%**

2019 – 2020

Projects

Deep Packet Inspection (DPI) Engine – Source Code C++, Multi Threading, Hashing

- Built a C++ DPI Engine that parses raw network packets across Ethernet/IP/TCP layers and extracts TLS SNI fields from HTTPS handshakes to classify and block applications like YouTube and Facebook, achieving latency of **<1ms per packet**.
- Implemented **stateful Five-Tuple flow tracking** with **HashMap**, enabling persistent block decisions across all packets of a connection without redundant re-inspection reducing **CPU overhead by 60%**.
- Architected a **multi-threaded pipeline** (Reader → Load Balancer → Fast Path → Output Writer) using **consistent hashing** for flow affinity and **thread-safe queues with mutex/condition variables**, improving throughput by **4× over single-threaded baseline across 4 worker threads**.
- Engineered a **rule-based policy engine** supporting IP blacklists, app-level filtering, and domain keyword matching, with per-application traffic classification reports on over 10000 packets.

Optimal Zone Discovery System – Live Demo React, Node.js, Supabase, Leaflet.js, PostgreSQL

- Built a full-stack geospatial platform recommending optimal residential zones using a custom designed **Radial Candidate Evaluation and Deduplication (RCED)** algorithm reducing candidate evaluation time by **70%** compared to brute-force spatial search.
- RCED** balances amenity diversity, walking-time buffers, and workplace proximity to return 5 distinct, high-utility neighborhood recommendations.
- Developed a **spatial sliding-window search** evaluating amenity density across geographic candidates within user-defined travel-time and category constraints.
- Applied **weighted multi-objective scoring** with geometric deduplication **filtering 80%+ overlapping cluster candidates** and improving recommendation precision.
- Deployed frontend on **Vercel**, backend on **Render**, and database on **Supabase** — production-grade cloud infrastructure.

Club Activity Management System – Live Demo Flask, MySQL, Jinja2

- Developed a full-stack web application enabling students to explore and manage 100+ clubs and activities through a dynamic database-driven interface.
- Designed **normalized relational schema with foreign key constraints** and automated event updates based on timestamps.
- Built scalable admin dashboard with **dynamic CRUD APIs**, cutting average content update time from **15 minutes to under 30 seconds** for non-technical administrators.
- Implemented secure **authentication, session handling, and RESTful routing** for secure, role-based access control enforcing 3-tier permission boundaries.
- Adopted **Agile SDLC across 4 iterative sprints**, enabling continuous testing at each phase and delivering a production-ready system 2 weeks ahead of schedule.

E-Commerce Platform with Customer Segmentation – Live Demo Flask, MySQL, K-Means Clustering

- Built scalable **REST APIs** using Flask with JWT-based authentication and transactional workflow management.
- Optimized an **ACID-compliant relational database** with indexed queries supporting multiple concurrent users.
- Applied **K-Means Clustering** over **10,000+ records** to generate insights on seasonal trends and customer purchasing patterns.
- Visualized sales trends and user behavior via interactive dashboards with a fully **responsive frontend**.

Technical Skills

Languages: Python, Java, C++, JavaScript

Backend: Node.js, Express.js, Flask, REST APIs, JWT Authentication, MVC

Frontend: HTML, CSS, Tailwind CSS, React

Databases: MySQL, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, Postman, VS Code, Supabase

Algorithmic Competitions & Achievements

- 3-Star Coder (Max Rating: 1700)** on [CodeChef](#).
- Guardian: (Rating: 2126)** on [LeetCode](#) with **1000+ problems solved**.
- Ranked 764** among **43,000+** participants in LeetCode Weekly Contest 490.
- Ranked 145** among **3,000+** participants in CodeChef Starters 220.
- Certification in **AWS - EC2 and S3**.